

Doug Morgen

Staff Software Engineer | AI Platforms | Financial Systems | Product Architecture

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PROFESSIONAL SUMMARY

Staff-level software engineer with 20+ years designing AI platforms, financial systems, ecommerce infrastructure, ERP software, payment platforms, and workflow automation. Former founder of an \$8M/year ecommerce company, Senior Software Engineer at EasyPost building production billing systems supporting billions in annual shipping volume, and architect of multiple end-to-end SaaS products.

Specialize in designing business-critical systems where correctness, extensibility, automation, and product thinking matter.

CORE EXPERTISE

Financial Systems

Payments • Billing • ERP • General Ledger • Payroll • Job Costing • Financial Reporting • Reconciliation • Commission Systems

Artificial Intelligence

LLMs • RAG • OCR • Semantic Search • Multimodal AI • Video Processing • AI Agents • Prompt Orchestration

Platform Engineering

Ruby on Rails • PostgreSQL • Redis • Sidekiq • React • TypeScript • Python • Docker • AWS • Heroku • GitHub Actions

PROFESSIONAL EXPERIENCE

HomeBrand - Principal Engineer / CTO

Architected an AI-native property management platform from concept through production as the lead engineer.

- Designed a multimodal AI platform combining OCR, document ingestion, semantic retrieval, embeddings, and LLM orchestration to answer questions from homeowner documents, receipts, warranties, manuals, and maintenance records.
- Built a video ingestion pipeline capable of synchronizing speech transcription, visual content, temporal segmentation, and semantic retrieval to transform long homeowner walkthrough videos into searchable knowledge.
- Designed retrieval and chunking strategies that dramatically reduced LLM context size while improving response quality, latency, and cost.
- Owned backend architecture, frontend development, infrastructure, deployment, and product architecture across the platform.

Arrafund - Principal Engineer / CTO

Architected the core platform for a creator economy payment and attribution system.

- Designed a generic transaction and event architecture that decoupled payment providers from business logic, enabling Stripe, Wise, commissions, payouts, and future payment providers to share a common extensible transaction model.
- Built a highly flexible attribution framework capable of assigning creator, referral, affiliate, and commerce attribution from arbitrary webhook events.
- Designed scalable webhook infrastructure supporting future integrations without architectural changes.
- Architected payment infrastructure emphasizing extensibility, correctness, and long-term scalability.

BrightGo - Senior Software Engineer

Partnered directly with the CTO to build payroll, ERP, and accounting systems.

- Owned approximately two-thirds of the accounting platform and half of the payroll platform while mentoring two Ruby on Rails engineers.
- Designed the job-costing architecture translating operational activity into payroll allocations and accounting ledger entries consumed by third-party payroll and accounting systems.
- Redesigned the chart of accounts and P&L reporting, enabling the ERP to produce financial statements matching customers' legacy accounting systems and helping close new enterprise business.
- Increased financial import throughput approximately 10× by redesigning ledger batching while maintaining accounting correctness and idempotent processing.
- Built financial reporting, reconciliation, and accounting infrastructure where correctness and auditability were mission-critical.

EasyPost - Senior Software Engineer

Architected production billing systems within a logistics platform supporting billions of dollars in annual shipping volume.

- Became the primary engineer responsible for EasyPost's legacy transaction and delta billing platform, reverse-engineering complex financial workflows that few engineers wanted to modify.
- Redesigned the Rails billing architecture to integrate with Python billing services responsible for carrier invoice calculations.
- Built financial systems capable of handling continuously changing carrier invoices while preserving billing correctness across multiple downstream adjustments.
- Developed production billing integrations supporting UPS, FedEx, DHL, and additional carrier billing workflows.
- Contributed to the evolution of the billing platform toward service-oriented Python microservices.

Founder / CTO

Cellular Country

Founded and scaled an \$8M/year ecommerce business while architecting the software platform powering company operations.

- Designed and built an integrated business platform supporting inventory management, purchasing, pricing, warehouse operations, fulfillment, CRM, accounting, and marketplace automation.
- Architected an automated reverse-logistics platform processing approximately 100 smartphones per day (~\$5M annual resale volume).
- Built a real-time inventory pipeline where scanning a single device automatically synchronized listings across the company website, eBay, and Amazon.
- Developed a dynamic pricing engine that continuously optimized resale prices based on sales velocity, balancing gross margin with inventory turnover.
- Built an automated auction bidding engine analyzing 10,000-line inventory feeds using pricing, inventory levels, expected resale value, quantities, and business constraints to optimize purchasing decisions.
- Automated warehouse operations including barcode scanning, IMEI tracking, shipping label generation, pick/pack workflows, fraud detection, and fulfillment.

EDUCATION

M.S. Mechanical Engineering
University of California, San Diego

B.S. Engineering Science & Mechanics
Virginia Tech